

用户登录测试

#1.用户登录并获取 token

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/user_login" \
-H "Content-Type: application/json" \
-d '{"email": "18165031775@qq.com", "passwd": "123456"}' \
-s | python3 -m json.tool
{
  "email": "18165031775@qq.com",
  "error": 0,
  "host": "39.102.149.84",
  "port": "8091",
  "token": "951211d0-26b5-478a-9616-e6580d6dc9f3",
  "uid": 1003
}
```

登录成功，获取 token: 951211d0-26b5-478a-9616-e6580d6dc9f3

记录下：

TOKEN="951211d0-26b5-478a-9616-e6580d6dc9f3"

UID=1003

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/user_login" \
-H "Content-Type: application/json" \
-d '{"uid": 1073, "password": "qwertyu1234@"}' \
-s | python3 -m json.tool
{
  "error": 1009
}
```

密码错误返回错误码

贴吧管理接口测试

1. 获取论坛列表

```
curl "http://39.102.149.84:8080/forums?page=1&limit=10" \
```

```
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/forums?page=1&limit=10" \
-s | python3 -m json.tool
{
  "error": 0,
  "forums": [
    {
      "avatar": "/res/forum_tech.jpg",
      "description": "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
      "follower_count": 2,
      "forum_id": 1,
      "name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "post_count": 10
    },
    {
      "avatar": "/res/forum_photo.jpg",
      "description": "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
      "follower_count": 3,
      "forum_id": 2,
      "name": "\u6444\u5f71\u4ea4\u6d41\u5427",
      "post_count": 1
    },
    {
      "avatar": "/res/forum_game.jpg",
      "description": "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
      "follower_count": 1,
      "forum_id": 3,
      "name": "\u6e38\u620f\u7ec4\u961f\u5427",
      "post_count": 0
    }
  ]
}
```

2. 获取论坛详情

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/forum/detail?forum_id=1&uid=$UID&token=$TOKEN" \
-s | python3 -m json.tool
{
  "error": 0,
  "forum": {
    "avatar": "/res/forum_tech.jpg",
    "description": "\u804a\u786c\u4ef6\u540e\u7aef",
    "follower_count": 2,
    "forum_id": 1,
    "is_followed": false,
    "name": "\u79d1\u6280\u8ba8\u8bba\u5427",
    "post_count": 10
  }
}
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/forum/detail?forum_id=2&uid=$UID&token=$TOKEN" -s | python3 -m json.tool
{
  "error": 0,
  "forum": {
    "avatar": "/res/forum_photo.jpg",
    "description": "\u5668\u6750\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
    "follower_count": 3,
    "forum_id": 2,
    "is_followed": false,
    "name": "\u6444\u5f71\u4ea4\u6d41\u5427",
    "post_count": 1
  }
}
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/forum/detail?forum_id=3&uid=$UID&token=$TOKEN" -s | python3 -m json.tool
{
  "error": 0,
  "forum": {
    "avatar": "/res/forum_game.jpg",
    "description": "\u5f00\u9ed1\u540e\u7565\u4e0e\u8054\u6ecb\u7ea6\u6218",
    "follower_count": 1,
    "forum_id": 3,
    "is_followed": false,
    "name": "\u6e38\u620f\u7ec4\u961f\u5427",
    "post_count": 0
  }
}
```

forum_id	name	avatar	description	owner_uid	follower_cnt	post
1	科技讨论吧	/res/forum_tech.jpg	聊硬件/AI/后端	1054	2	
2	摄影交流吧	/res/forum_photo.jpg	器材、后期与旅拍分享	1055	3	
3	游戏组队吧	/res/forum_game.jpg	开黑、攻略与联机约战	1056	1	

3. 关注贴吧

```
curl -X POST "http://39.102.149.84:8080/follow_forum" \
-H "Content-Type: application/json" \
-d '{"uid": 1003, "token": "\u951211d0-26b5-478a-9616-e6580d6dc9f3", "forum_id": 1}' \
-s | python3 -m json.tool
```

校验失败

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/follow_forum" \
-H "Content-Type: application/json" \
-d '{"uid": $UID, "token": "$TOKEN", "forum_id": 1}' \
-s | python3 -m json.tool
{
  "error": 1010
}
```

校验成功

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/follow_forum" \
-H "Content-Type: application/json" \
-d '{"uid": 1003, "token": "\u951211d0-26b5-478a-9616-e6580d6dc9f3", "forum_id": 1}' \
-s | python3 -m json.tool
{
  "error": 0
}
```

发现数据库中关注人数加 1

1	1 科技讨论吧	/res/forum_tech.jpg	聊硬件/AI/后端	1054	2	10 2025-11-23 23:35:39	2025-12-02 16:24:29
---	---------	---------------------	-----------	------	---	------------------------	---------------------

4. 取消关注贴吧

```
curl -X POST "http://39.102.149.84:8080/unfollow_forum" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", \  
"forum_id": 1}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/unfollow_forum" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", "forum_id": 1}' \  
-s | python3 -m json.tool \  
{ \  
  "error": 1010 \  
} \  
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/unfollow_forum" -H "Content-Type: application/json" -d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", "forum_id": 1}' -s | python3 -m json.tool \  
{ \  
  "error": 0 \  
} \  
usr@usr-VMware-Virtual-Platform:~$
```

同理验证取消关注，数据库中关注量减一

1	1 科技讨论吧	/res/forum_tech.jpg	聊硬件/AI/后端	1054	1	10 2025-11-23 23:35:39	2025-12-02 16:26:21
---	---------	---------------------	-----------	------	---	------------------------	---------------------

5. 获取已关注的贴吧列表

```
curl "http://39.102.149.84:8080/followed_forums?uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3&page=1&limit=10" \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/followed_forums?uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3&page=1&limit=10" -s | python3 -m json.tool \  
{ \  
  "error": 0 \  
} \  
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/follow_forum" -H "Content-Type: application/json" -d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", "forum_id": 1}' -s | python3 -m json.tool \  
{ \  
  "error": 0 \  
} \  
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/followed_forums?uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3&page=1&limit=10" -s | python3 -m json.tool \  
{ \  
  "error": 0, \  
  "forums": [ \  
    { \  
      "avatar": "/res/forum_tech.jpg", \  
      "description": "\u004a\u786c\u4ef6/AI/\u540e\u7aef", \  
      "follower_count": 3, \  
      "forum_id": 1, \  
      "name": "\u79d1\u6280\u8ba8\u8bba\u533a", \  
      "post_count": 10 \  
    } \  
  ] \  
} \  
usr@usr-VMware-Virtual-Platform:~$
```

分别在有关关注和没关注的情况下查询关注列表

帖子管理接口测试

1. 获取帖子列表 (可按不同方式排序)

按时间排序

```
curl "http://39.102.149.84:8080/posts?forum_id=1&page=1&limit=10&sort=time" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/posts?forum_id=1&page=1&limit=10&sort=time" \
-s | python3 -m json.tool
{
  "error": 0,
  "posts": [
    {
      "author": "1120221775",
      "content_preview": "\u5965\u8428\u8482\u5965\u8428\u8482\u554a\u554a\u554a",
      "created_at": "2025-12-02 16:24:29",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 13,
      "reply_count": 0,
      "title": "\u5965\u8428\u8482",
      "uid": 1063
    },
    {
      "author": "1120221775",
      "content_preview": "\u6492\u6253\u53d1\u6253\u53d1",
      "created_at": "2025-12-01 00:22:53",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 12,
      "reply_count": 0,
      "title": "SFDGDS\u53d1",
      "uid": 1063
    },
    {
      "author": "1120221775",
      "content_preview": "\u8428\u8fbc",
      "created_at": "2025-11-29 16:21:38",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 11,
      "reply_count": 1,
      "title": "\u5176\u5b9e\u6211\u7684",
      "uid": 1063
    }
  ]
}
```

按热度排序

```
curl "http://39.102.149.84:8080/posts?forum_id=1&page=1&limit=10&sort=hot" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/posts?forum_id=1&page=1&limit=10&sort=hot"
-s | python3 -m json.tool
{
  "error": 0,
  "posts": [
    {
      "author": "sqy",
      "content_preview": "\u6b63\u6587\u5185\u5bb9",
      "created_at": "2025-11-24 00:28:23",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 4,
      "reply_count": 7,
      "title": "\u6d4b\u8bd5\u53d1\u5e16",
      "uid": 1054
    },
    {
      "author": "1120221775",
      "content_preview": "\u8428\u8fbc",
      "created_at": "2025-11-29 16:21:38",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 11,
      "reply_count": 1,
      "title": "\u5176\u5b9e\u6211\u7684",
      "uid": 1063
    },
    {
      "author": "ocz",
      "content_preview": "\u804a\u804a\u4f60\u7684\u4e3b\u673a/\u914d\u7f6e",
      "created_at": "2025-11-24 00:00:46",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 2,
      "reply_count": 0,
      "title": "\u786c\u4ef6\u914d\u7f6e\u4ea4\u6d41",
      "uid": 1055
    },
    {
      "author": "1120221775",
      "content_preview": "\u5965\u8428\u8482\u5965\u8428\u8482\u5554\u5554\u5554",
      "created_at": "2025-12-02 16:24:29",
      "forum_id": 1,

```

2. 创建帖子

```
curl -X POST "http://39.102.149.84:8080/post/create" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"forum_id\": 1, \"title\": \"接口测试用帖\", \"content\": \"阿马特拉斯\"}" \
-s | python3 -m json.tool
```

```
usr@usr-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/post/create" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"forum_id\": 1, \"title\": \"接口测试用帖\", \"content\": \"阿马特拉斯\"}" \
-s | python3 -m json.tool
{
  "error": 0,
  "post_id": 14
}
```

14	14	1	1003	接口测试用帖	阿马特拉斯	0
----	----	---	------	--------	-------	---

Post_id=14

3. 获取帖子详情

```
curl "http://39.102.149.84:8080/post/detail?post_id=1&uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3" \
-s | python3 -m json.tool
```

```
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/post/detail?post_id=14&uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3" \
-s | python3 -m json.tool
{
  "error": 0,
  "post": {
    "author": "tc",
    "collect_count": 0,
    "content": "\u963f\u9a6c\u7279\u62c9\u65af",
    "created_at": "2025-12-02 16:52:23",
    "forum_id": 1,
    "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
    "is_collected": false,
    "is_essence": false,
    "is_liked": false,
    "is_top": false,
    "like_count": 0,
    "post_id": 14,
    "reply_count": 0,
    "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16",
    "uid": 1003
  }
}
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/post/detail?post_id=5&uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3" -s | python3 -m json.tool
{
  "error": 0,
  "post": {
    "author": "sqy",
    "collect_count": 0,
    "content": "\u66b3\u6587\u5185\u5bb9",
    "created_at": "2025-11-24 00:35:46",
    "forum_id": 1,
    "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
    "is_collected": false,
    "is_essence": false,
    "is_liked": false,
    "is_top": false,
    "like_count": 0,
    "post_id": 5,
    "reply_count": 0,
    "title": "\u6d4b\u8bd5\u53d1\u5e16",
    "uid": 1054
  }
}
```

分别测试自己的和他人的帖子

4. 更新帖子

```
curl -X POST "http://39.102.149.84:8080/post/update" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": $POST_ID, \"title\": \"更新后的标题\", \"content\": \"更新后的内容\"}" \
-s | python3 -m json.tool
```

```
usr@usr-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/post/update" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": 14, \"title\": \"更新后的标题\", \"content\": \"更新后的内容\"}" \
-s | python3 -m json.tool
{
  "error": 0
}
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/post/detail?post_id=14&uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3" -s | python3 -m json.tool
{
  "error": 0,
  "post": {
    "author": "tc",
    "collect_count": 0,
    "content": "\u66f4\u65b0\u540e\u7684\u5185\u5bb9",
    "created_at": "2025-12-02 16:52:23",
    "forum_id": 1,
    "forum_name": "\u79d1\u6280\u8ba8\u8ba8",
    "is_collected": false,
    "is_essence": false,
    "is_liked": false,
    "is_top": false,
    "like_count": 0,
    "post_id": 14,
    "reply_count": 0,
    "title": "\u66f4\u65b0\u540e\u7684\u5185\u5bb9",
    "uid": 1003
  }
}
```

--	--	-	----	~~~~	~~~~	~~~~	~~~~
14	14	1	1003	更新后的标题	更新后的内容	0	

5. 删除帖子 (软删除)

```
curl -X POST "http://39.102.149.84:8080/post/delete" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": $POST_ID}" \
-s | python3 -m json.tool
```

```
usr@usr-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/post/delete" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": 14}" \
-s | python3 -m json.tool
{
  "error": 0
}
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/post/detail?post_id=14&uid=1003&token=951211d0-26b5-478a-9616-e6580d6dc9f3" -s | python3 -m json.tool
{
  "error": 1016
}
```

id	forum_id	uid	title	content	is_top	is_essence	like_cnt	collect_cnt	reply_cnt	status	create_at
1	1	1054	新标题	新正文	0	0	0	0	0	0	2025-11-
2	1	1055	硬件配置交流	聊聊你的主机/配置	0	0	1	1	0	1	2025-11-
3	2	1054	新标题	新正文	0	0	0	0	0	1	2025-11-
4	1	1054	测试发帖	正文内容	0	0	1	2	7	1	2025-11-
5	1	1054	测试发帖	正文内容	0	0	0	0	0	1	2025-11-
6	1	1063	asd	cfdfd	0	0	0	0	0	1	2025-11-
7	1	1063	撒杜甫	萨法是的啊三大法	0	0	0	0	0	0	2025-11-
8	1	1063	色达是的	萨法是的	0	0	0	0	0	1	2025-11-
9	1	1063	色达是的	萨法是的	0	0	0	0	0	1	2025-11-
10	1	1063	其实我的	萨达	0	0	0	0	0	1	2025-11-
11	1	1063	其实我的	萨达	0	0	0	0	1	1	2025-11-
12	1	1063	SFDGDS发	撒打发打发	0	0	0	0	0	1	2025-12-
13	1	1063	奥萨蒂	奥萨蒂奥萨蒂啊啊	0	0	0	0	0	1	2025-12-
14	1	1003	更新后的标题	更新后的内容	0	0	0	0	0	0	2025-12-

6. 点赞/取消点赞帖子

```
curl -X POST "http://39.102.149.84:8080/like_post" \  
-H "Content-Type: application/json" \  
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \  
\"post_id\": 15}" \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/like_post" \  
-H "Content-Type: application/json" \  
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": 15}" \  
-s | python3 -m json.tool  
{  
  "error": 0  
}
```

15	1	1003	接口测试用帖	阿马特拉斯	0	0	1
----	---	------	--------	-------	---	---	---

```
curl -X POST "http://39.102.149.84:8080/unlike_post" \  
-H "Content-Type: application/json" \  
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \  
\"post_id\": 15}" \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/unlike_post" \  
-H "Content-Type: application/json" \  
-d "{\"uid\": 1003, \"token\": \"951211d0-26b5-478a-9616-e6580d6dc9f3\", \"post_id\": 15}" \  
-s | python3 -m json.tool  
{  
  "error": 0  
}
```

15	1	1003	接口测试用帖	阿马特拉斯	0	0	0
----	---	------	--------	-------	---	---	---

7. 收藏/取消收藏帖子

```
curl -X POST "http://39.102.149.84:8080/collect_post" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3",  
"post_id": 15}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/collect_post" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", "post_id": 15}' \  
-s | python3 -m json.tool  
{  
  "error": 0  
}  
usr@usr-VMware-Virtual-Platform:~$
```

st_id	forum_id	uid	title	content	is_top	is_essence	like_cnt	collect_cnt
1	1	1054	新标题	新正文	0	0	0	0
2	1	1055	硬件配置交流	聊聊你的主机/配置	0	0	1	1
3	2	1054	新标题	新正文	0	0	0	0
4	1	1054	测试发帖	正文内容	0	0	1	2
5	1	1054	测试发帖	正文内容	0	0	0	0
6	1	1063	asd	cfdfd	0	0	0	0
7	1	1063	撒杜甫	萨法是的啊三大法	0	0	0	0
8	1	1063	色达是的	萨法是的	0	0	0	0
9	1	1063	色达是的	萨法是的	0	0	0	0
10	1	1063	其实我的	萨达	0	0	0	0
11	1	1063	其实我的	萨达	0	0	0	0
12	1	1063	SFDGDS发	撒打发打发	0	0	0	0
13	1	1063	奥萨蒂	奥萨蒂奥萨蒂啊啊	0	0	0	0
14	1	1003	更新后的标题	更新后的内容	0	0	0	0
15	1	1003	接口测试用帖	阿马特拉斯	0	0	0	1

```
curl -X POST "http://39.102.149.84:8080/uncollect_post" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3",  
"post_id": 15}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/uncollect_post" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "951211d0-26b5-478a-9616-e6580d6dc9f3", "post_id": 15}' \  
-s | python3 -m json.tool  
{  
  "error": 0  
}  
usr@usr-VMware-Virtual-Platform:~$
```

5	15	1	1003	接口测试用帖	阿马特拉斯	0	0	0	0
---	----	---	------	--------	-------	---	---	---	---

8. 置顶/取消置顶帖子

(需要贴吧 owner 权限) //此后的 token 改为 d6b4c8fd-e36b-401c-91e4-b6e27b37253e
换号, UID=1054, TOKEN= 3a800ff1-0be2-4258-9443-14b3f9e9d1da

```
curl -X POST "http://39.102.149.84:8080/set_top_post" \  
-H "Content-Type: application/json" \  
-d '{"uid\': 1054, \"token\': \"3a800ff1-0be2-4258-9443-14b3f9e9d1da\",  
\"post_id\': 16, \"is_top\': true}" \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/set_top_post" -H "Content-Type: application/json" -d '{"uid\': 1054, \"token\': \"3a800ff1-0be2-4258-9443-14b3f9e9d1da\", \"post_id\': 16, \"is_top\': true}" -s | python3 -m json.tool  
{  
  "error": 0  
}
```

9. 加精/取消加精帖子

(需要贴吧 owner 权限)

```
curl -X POST "http://39.102.149.84:8080/set_essence_post" \  
-H "Content-Type: application/json" \  
-d '{"uid\': 1054, \"token\': \"3a800ff1-0be2-4258-9443-14b3f9e9d1da\",  
\"post_id\': 16, \"is_essence\': true}" \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/set_essence_post" \  
-H "Content-Type: application/json" \  
-d '{"uid\': 1054, \"token\': \"3a800ff1-0be2-4258-9443-14b3f9e9d1da\", \"post_id\': 16, \"is_essence\': true}" -s | python3 -m json.tool  
{  
  "error": 0  
}
```

结果如下:

WHERE							
	post_id	forum_id	uid	title	content	is_top	is_essence
1	1	1	1054	新标题	新正文	0	0
2	2	1	1055	硬件配置交流	聊聊你的主机/配置	0	0
3	3	2	1054	新标题	新正文	0	0
4	4	1	1054	测试发帖	正文内容	0	0
5	5	1	1054	测试发帖	正文内容	0	0
6	6	1	1063	asd	cfdfd	0	0
7	7	1	1063	撒杜甫	萨法是的啊三大法宝v	0	0
8	8	1	1063	色达是的	萨法是的	0	0
9	9	1	1063	色达是的	萨法是的	0	0
10	10	1	1063	其实我的	萨达	0	0
11	11	1	1063	其实我的	萨达	0	0
12	12	1	1063	SFD6DS发	撒打发打发	0	0
13	13	1	1063	奥萨蒂	奥萨蒂奥萨蒂啊啊啊	0	0
14	14	1	1003	更新后的标题	更新后的内容	0	0
15	15	1	1003	接口测试用帖	阿马特拉斯	0	0
16	16	1	1003	神经病发作帖子	计算机男一辈子应该谈四个女朋友第一个女朋友,是计组。她总能把冰冷的	1	1
17	17	2	1003	接口测试用帖	阿马特拉斯	0	0
18	18	3	1003	接口测试用帖	阿马特拉斯	0	0

回复管理接口测试

1. 获取帖子回复列表

curl

```
"http://39.102.149.84:8080/post/replies?post_id=4&page=1&limit=10&sort=time&uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e" \
```

-s | python3 -m json.tool

```
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/post/replies?post_id=4&page=1&limit=10&sort=time&uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e" \
-s | python3 -m json.tool
{
  "error": 0,
  "replies": [
    {
      "author": "1120221775",
      "content": "\u539f\u6765\u5982\u6b64",
      "created_at": "2025-12-02 14:37:59",
      "floor": 3,
      "is_liked": false,
      "like_count": 0,
      "parent_reply_id": 0,
      "post_id": 4,
      "reply_id": 10,
      "uid": 1063
    },
    {
      "author": "1120221775",
      "content": "\u5965\u8428\u8482",
      "created_at": "2025-12-02 01:21:31",
      "floor": 2,
      "is_liked": false,
      "like_count": 1,
      "parent_reply_id": 0,
      "post_id": 4,
      "reply_id": 8,
      "uid": 1063
    },
    {
      "author": "uioojhyu",
      "content": "\u8fd9\u662f\u4e00\u6761\u6d4b\u8bd5\u56de\u590d",
      "created_at": "2025-11-29 18:55:58",
      "floor": 1,
      "is_liked": false,
      "like_count": 1,
      "parent_reply_id": 0,
      "post_id": 4,
      "reply_id": 3,
      "sub_replies": [
        {
          "author": "uioojhyu",
          "content": "\u6d4b\u8bd5",
          "created_at": "2025-12-01 15:12:34",
          "floor": 0,
          "is_liked": false,
```

2. 创建回复 (主回复, parent_reply_id=0)

```
curl -X POST "http://39.102.149.84:8080/reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\",
\"post_id\": 15, \"content\": \"ciao~\", \"parent_reply_id\": 0}" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/reply" -H "Content-Type: application/json" -d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\", \"post_id\": 15, \"content\": \"ciao~\", \"parent_reply_id\": 0}" -s | python3 -m json.tool
{
  "error": 0,
  "reply_id": 12
}
usr@usr-VMware-Virtual-Platform:~$
```

记录返回的 reply_id

REPLY_ID=12

回复得奇葩一点也没问题

```
curl -X POST "http://39.102.149.84:8080/reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\",
\"post_id\": 15, \"content\": \" Ciallo~(∠ · ω< )∩☆\", \"parent_reply_id\":
0}" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\", \"post_id\": 15, \"content\": \" Ciallo~(∠ · ω< )∩☆\", \"parent_reply_id\": 0}" -s | python3 -m json.tool
{
  "error": 0,
  "reply_id": 17
}
usr@usr-VMware-Virtual-Platform:~$
```

3. 创建楼中楼回复 (回复某个回复)

```
curl -X POST "http://39.102.149.84:8080/reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\",
\"post_id\": 15, \"content\": \"sixsixsixhaiyouqianlianwanhua\",
\"parent_reply_id\": 12}" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\", \"post_id\": 15, \"content\": \"sixsixsixhaiyouqianlianwanhua\", \"parent_reply_id\": 12}" -s | python3 -m json.tool
{
  "error": 0,
  "reply_id": 13
}
usr@usr-VMware-Virtual-Platform:~$
```

4. 删除回复

```
curl -X POST "http://39.102.149.84:8080/reply/delete" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "d6b4c8fd-e36b-401c-91e4-b6e27b37253e",  
"reply_id": 12}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/reply/delete" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "d6b4c8fd-e36b-401c-91e4-b6e27b37253e", "reply_id": 12}' \  
-s | python3 -m json.tool  
{  
  "error": 0  
}
```

12	12	15	1003 ciallo~	0	0	2	0	0
----	----	----	--------------	---	---	---	---	---

5. 点赞/取消点赞回复

```
curl -X POST "http://39.102.149.84:8080/like_reply" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "d6b4c8fd-e36b-401c-91e4-b6e27b37253e",  
"reply_id": 17}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/like_reply" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "d6b4c8fd-e36b-401c-91e4-b6e27b37253e", "reply_id": 17}' \  
-s | python3 -m json.tool  
{  
  "error": 0  
}  
usr@usr-VMware-Virtual-Platform:~$
```

WHERE		ORDER BY							
reply_id	post_id	uid	content	parent_reply_id	root_reply_id	floor	like_cnt	st	
1	4	1054	这是顶层回复	0	0	1	0	0	
2	4	1054	楼中楼	1	1	0	0	0	
3	4	1060	这是一条测试回复	0	0	1	1	1	
4	4	1060	这是一条测试回复	0	0	2	0	0	
5	11	1060	12/1 666这个入,	0	0	1	1	1	
6	4	1060	测试2	3	3	0	1	1	
7	4	1060	测试3	6	3	0	0	0	
8	4	1063	奥萨蒂	0	0	2	1	1	
9	4	1063	奥萨蒂啊实打实的	7	3	0	0	0	
10	4	1063	原来如此	0	0	3	0	0	
11	15	1003	ciallo~	0	0	1	0	0	
12	15	1003	ciallo~	0	0	2	0	0	
13	15	1003	sixsixsixhaiy...	12	12	0	0	0	
14	15	1063	@tc 啊啊啊	13	12	0	0	0	
15	4	1063	@vioojhyu 哈哈呀	6	3	0	0	0	
16	4	1063	@1120221775 奥..	10	10	0	0	0	
17	15	1003	Ciallo~(ノ・ω・	0	0	2	1	1	

```
curl -X POST "http://39.102.149.84:8080/unlike_reply" \  
-H "Content-Type: application/json" \  
-d '{"uid": 1003, "token": "d6b4c8fd-e36b-401c-91e4-b6e27b37253e",  
"reply_id": 17}' \  
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl -X POST "http://39.102.149.84:8080/unlike_reply" \
-H "Content-Type: application/json" \
-d "{\"uid\": 1003, \"token\": \"d6b4c8fd-e36b-401c-91e4-b6e27b37253e\", \"reply_id\": 17}" \
-s | python3 -m json.tool
{
  "error": 0
}
```

17	15	1003	Ciallo~(ノ・ω・	0	0	2	0
----	----	------	--------------	---	---	---	---

搜索功能接口测试

1. 搜索帖子

```
curl "http://39.102.149.84:8080/search?keyword=接口测试用帖
&type=post&page=1&limit=10" \
-s | python3 -m json.tool
```

提前在三个吧里都设置了同样标题的帖子

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=接口测试用帖&type=post&page=1&limit=10" -s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 20:08:37",
      "forum_id": 3,
      "forum_name": "\u6e38\u620f\u7ec4\u961f\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 18,
      "reply_count": 0,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    },
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 20:08:27",
      "forum_id": 2,
      "forum_name": "\u6444\u5f71\u4ea4\u6d41\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 17,
      "reply_count": 0,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    },
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 19:05:24",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 15,
      "reply_count": 7,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    }
  ]
}
```

15	1	1003	接口测试用帖	阿马特拉斯	
16	1	1003	神经病发作帖子	计算机男一辈子应该谈四个女朋友第一个女朋友，是计组。她总能把冰冷的	
17	2	1003	接口测试用帖	阿马特拉斯	
18	3	1003	接口测试用帖	阿马特拉斯	

也可以只有关键词匹配

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=接口&type=post&page=1&limit=10" -s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 20:08:37",
      "forum_id": 3,
      "forum_name": "\u6e38\u620f\u7ec4\u961f\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 18,
      "reply_count": 0,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    },
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 20:08:27",
      "forum_id": 2,
      "forum_name": "\u6444\u5f71\u4ea4\u6d41\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 17,
      "reply_count": 0,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    },
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 19:05:24",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 15,
      "reply_count": 7,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    }
  ]
}
```

2. 在指定贴吧内搜索帖子

```
curl "http://39.102.149.84:8080/search?keyword= 接 口 测 试 用 帖
&type=post&forum_id=1&page=1&limit=10" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=接口测试用帖&type=post&forum_id=1&page=1&limit=10" -s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 19:05:24",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 15,
      "reply_count": 7,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    }
  ]
}
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=接口测试用帖&type=post&forum_id=3&page=1&limit=10" -s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 20:08:37",
      "forum_id": 3,
      "forum_name": "\u6e38\u620f\u7ec4\u961f\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 0,
      "post_id": 18,
      "reply_count": 0,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    }
  ]
}
usr@usr-VMware-Virtual-Platform:~$
```


3. 搜索贴吧

```
curl "http://39.102.149.84:8080/search?keyword= 游 戏 组 队 吧
&type=forum&page=1&limit=10" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=游戏组队吧&type=forum&page=1&limit=10" \
-s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "avatar": "/res/forum_game.jpg",
      "description": "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
      "follower_count": 1,
      "forum_id": 3,
      "name": "\u6e38\u620f\u7ec4\u961f\u5427",
      "post_count": 1
    }
  ]
}
```

4. 搜索用户

```
curl
"http://39.102.149.84:8080/search?keyword=1120221775&type=user&page=1&limit=10" \
-s | python3 -m json.tool
```

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=1120221775&type=user&page=1&limit=10" -s | python3 -m json.tool
{
  "error": 0,
  "results": [
    {
      "email": "juffhhvh@163.com",
      "icon": "",
      "name": "1120221775",
      "nick": "1120221775",
      "sex": 0,
      "uid": 1063
    }
  ]
}
```

5. 空关键词测试

```
curl "http://39.102.149.84:8080/search?keyword=&type=post&page=1&limit=10" \
-s | python3 -m json.tool
```

产生空 json 错误码

```
usr@usr-VMware-Virtual-Platform:~$ curl "http://39.102.149.84:8080/search?keyword=&type=post&page=1&limit=10" \
-s | python3 -m json.tool
{
  "error": 1001
}
```

用户内容管理接口测试 s

1. 获取用户发布的帖子

```
curl "http://39.102.149.84:8080/my/posts?uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e&page=1&limit=10" \
-s | python3 -m json.tool
```

[illegible]

2. 获取用户发布的回复

```
curl "http://39.102.149.84:8080/my/replies?uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e&page=1&limit=10" \
-s | python3 -m json.tool
```

```
usr@usr-VWare-Virtual-Platform:~$ curl "http://39.102.149.84:8080/my/replies?uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e&page=1&limit=10" \
-s | python3 -m json.tool
{
  "error": 0,
  "replies": [
    {
      "content": "sixsixsixhaiyouqianlianwanhua",
      "created_at": "2025-12-02 19:29:29",
      "floor": 0,
      "like_count": 0,
      "parent_reply_id": 12,
      "post_id": 15,
      "post_title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "reply_id": 13,
      "root_reply_id": 12,
      "uid": 1003
    },
    {
      "content": "ciao~",
      "created_at": "2025-12-02 19:27:45",
      "floor": 1,
      "like_count": 0,
      "parent_reply_id": 0,
      "post_id": 15,
      "post_title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "reply_id": 11,
      "root_reply_id": 0,
      "uid": 1003
    }
  ]
}
```

3. 获取用户收藏的帖子

```
curl "http://39.102.149.84:8080/my/collections?uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e&page=1&limit=10" \
```

```
-s | python3 -m json.tool
```

```
usr@usr-Virtual-Platform:~$ curl "http://39.102.149.84:8080/my/collections?uid=1003&token=d6b4c8fd-e36b-401c-91e4-b6e27b37253e&page=1&limit=10" \
-s | python3 -m json.tool
{
  "error": 0,
  "posts": [
    {
      "author": "tc",
      "content_preview": "\u963f\u9a6c\u7279\u62c9\u65af",
      "created_at": "2025-12-02 19:05:24",
      "forum_id": 1,
      "forum_name": "\u79d1\u6280\u8ba8\u8bba\u5427",
      "is_essence": false,
      "is_top": false,
      "like_count": 1,
      "post_id": 15,
      "reply_count": 7,
      "title": "\u63a5\u53e3\u6d4b\u8bd5\u7528\u5e16 ",
      "uid": 1003
    }
  ]
}
```

错误码检验

```
enum ErrorCodes {
    Success = 0,
    Error_Json = 1001,    // Json 解析失败
    RPCFailed = 1002,    // RPC 调用失败
    VarifyExpired = 1003, // 验证码过期
    VarifyCodeErr = 1004, // 验证码错误
    UserExist = 1005,     // 用户已存在
    PasswdErr = 1006,     // 密码错误
    EmailNotMatch = 1007, // 邮箱不匹配
    PasswdUpFailed = 1008, // 更新密码失败
    PasswdInvalid = 1009, // 密码无效
    TokenInvalid = 1010,  // Token 失效
    UidInvalid = 1011,    // uid 无效
    ForumNotFound = 1012, // 贴吧不存在
    AlreadyFollowed = 1013, // 已关注
    NotFollowed = 1014,   // 未关注
    DbError = 1015,       // 数据库错误
    PostNotFound = 1016,  // 帖子不存在
    NoPermission = 1017,  // 无权限
    AlreadyLiked = 1018,  // 已点赞
    NotLiked = 1019,     // 未点赞
    AlreadyCollected = 1020, // 已收藏
    NotCollected = 1021,  // 未收藏
    ReplyNotFound = 1022,  // 回复不存在
    AlreadyLikedReply = 1023, // 回复已点赞
    NotLikedReply = 1024   // 回复未点赞
};
```

由于基础功能已于之前测试成功，因此错误码测试采用脚本进行
脚本如下：test_error_codes.sh

```
SERVER="http://39.102.149.84:8080"
LOG_FILE="error_test_$(date +%Y%m%d_%H%M%S).log"

echo "=== HTTP 接口错误码测试 ===" | tee -a $LOG_FILE
echo "测试时间: $(date)" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 记录函数
log_test() {
    echo "[$(date '+%H:%M:%S')] $1" | tee -a $LOG_FILE
}

test_error_case() {
    local test_name="$1"
    local method="$2"
    local endpoint="$3"
    local data="$4"
    local expected_code="$5"

    log_test "测试: $test_name (期望: $expected_code)"

    if [ "$method" = "GET" ]; then
        response=$(curl -s "$SERVER$endpoint")
    else
        response=$(curl -s -X $method -H "Content-Type: application/json" -d "$data"
"$SERVER$endpoint")
    fi

    # 解析错误码
    actual_code=$(echo "$response" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    print(data.get('error', -1))
except:
    print(-2)
" 2>/dev/null)

    log_test "响应: $response"

    if [ "$actual_code" = "$expected_code" ]; then
```

```

        log_test "☑ 通过: 期望 $expected_code, 实际 $actual_code"
        return 0
    else
        log_test "✗ 失败: 期望 $expected_code, 实际 $actual_code"
        return 1
    fi
}

echo ""
echo "1. 测试 JSON 解析错误 (1001)" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 1.1 无效的 JSON 格式
test_error_case "无效 JSON 格式" "POST" "/user_login" '{"invalid json' "1001"

# 1.2 缺少引号
test_error_case "缺少引号" "POST" "/user_login" '{email: "test@test.com", passwd: "123"}'
"1001"

# 1.3 多余的逗号
test_error_case "多余逗号" "POST" "/user_login" '{"email": "test@test.com", "passwd":
"123",}' "1001"

echo ""
echo "2. 测试用户登录相关错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 2.1 不存在的邮箱 (应该是 1009 或 1011)
test_error_case "不存在的邮箱" "POST" "/user_login" '{"email": "nonexistent@example.com",
"passwd": "123456"}' "1009"

# 2.2 错误密码
test_error_case "错误密码" "POST" "/user_login" '{"email": "18165031775@qq.com", "passwd":
"wrongpassword"}' "1009"

# 2.3 缺少必填字段
test_error_case "缺少 email" "POST" "/user_login" '{"passwd": "123456"}' "1009"
test_error_case "缺少 password" "POST" "/user_login" '{"email": "test@test.com"}' "1009"

# 2.4 空字段
test_error_case "空邮箱" "POST" "/user_login" '{"email": "", "passwd": "123456"}' "1009"
test_error_case "空密码" "POST" "/user_login" '{"email": "test@test.com", "passwd": ""}'
"1009"

echo ""

```

```

echo "3. 测试贴吧相关错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 先登录获取一个有效 token 用于测试
log_test "登录获取有效 token..."
LOGIN_RESPONSE=$(curl -s -X POST "$SERVER/user_login" \
    -H "Content-Type: application/json" \
    -d '{"email": "18165031775@qq.com", "passwd": "123456"}')

VALID_TOKEN=$(echo "$LOGIN_RESPONSE" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    print(data.get('token', ''))
except:
    print('')
")

VALID_UID=$(echo "$LOGIN_RESPONSE" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    print(data.get('uid', ''))
except:
    print('')
")

if [ -n "$VALID_TOKEN" ]; then
    log_test "获取到有效 token: ${VALID_TOKEN:0:20}..."

    # 3.1 不存在的贴吧 (1012)
    test_error_case "不存在的贴吧 ID" "GET"
"/forum/detail?forum_id=999999&uid=$VALID_UID&token=$VALID_TOKEN" "" "1012"

    # 3.2 无效的贴吧 ID 格式
    test_error_case "非数字贴吧 ID" "GET"
"/forum/detail?forum_id=abc&uid=$VALID_UID&token=$VALID_TOKEN" "" "1001"

    # 3.3 负数的贴吧 ID
    test_error_case "负数贴吧 ID" "GET" "/forum/detail?forum_id=-
1&uid=$VALID_UID&token=$VALID_TOKEN" "" "1012"

    # 3.4 重复关注 (先关注, 再关注一次)
    log_test "测试重复关注..."
    # 先关注一个存在的贴吧

```

```

curl -s -X POST "$SERVER/follow_forum" \
-H "Content-Type: application/json" \
-d "{\"uid\": $VALID_UID, \"token\": \"$VALID_TOKEN\", \"forum_id\": 1}" > /dev/null

# 再关注一次（应该成功，但我们可以检查数据库是否处理了重复）
test_error_case "重复关注" "POST" "/follow_forum" "{\"uid\": $VALID_UID, \"token\": \"$VALID_TOKEN\", \"forum_id\": 1}" "1013"

# 3.5 关注不存在的贴吧
test_error_case "关注不存在的贴吧" "POST" "/follow_forum" "{\"uid\": $VALID_UID, \"token\": \"$VALID_TOKEN\", \"forum_id\": 999}" "1012"

# 3.6 取消关注未关注的贴吧
test_error_case "取消未关注的贴吧" "POST" "/unfollow_forum" "{\"uid\": $VALID_UID, \"token\": \"$VALID_TOKEN\", \"forum_id\": 3}" "1014"

else
    log_test "✗ 无法获取有效 token，跳过贴吧测试"
fi

echo ""
echo "4. 测试 Token 相关错误 (1010)" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 4.1 无效 Token
test_error_case "无效 Token" "POST" "/follow_forum" '{"uid": 1001, "token": "invalid_token_123", "forum_id": 1}' "1010"

# 4.2 空 Token
test_error_case "空 Token" "POST" "/follow_forum" '{"uid": 1001, "token": "", "forum_id": 1}' "1001"

# 4.3 Token 与 UID 不匹配
if [ -n "$VALID_TOKEN" ]; then
    test_error_case "Token 与 UID 不匹配" "POST" "/follow_forum" "{\"uid\": 999999, \"token\": \"$VALID_TOKEN\", \"forum_id\": 1}" "1010"
fi

# 4.4 过期的 Token（需要模拟 token 过期）
# 暂时跳过，因为需要知道 token 过期机制

echo ""
echo "5. 测试帖子相关错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

```



```

if [ -n "$VALID_TOKEN" ]; then
    # 5.1 获取不存在的帖子详情 (1016)
    test_error_case "不存在的帖子" "GET"
"/post/detail?post_id=999999&uid=$VALID_UID&token=$VALID_TOKEN" "" "1016"

    # 5.2 在不存在贴吧发帖
    test_error_case "在不存在贴吧发帖" "POST" "/post/create" "{\"uid\": $VALID_UID,
\"token\": \"$VALID_TOKEN\", \"forum_id\": 999999, \"title\": \"测试\", \"content\": \"测试
内容\"}" "1012"

    # 5.3 更新不存在的帖子
    test_error_case "更新不存在的帖子" "POST" "/post/update" "{\"uid\": $VALID_UID,
\"token\": \"$VALID_TOKEN\", \"post_id\": 999999, \"title\": \"新标题\", \"content\": \"新内
容\"}" "1016"

    # 5.4 删除不存在的帖子
    test_error_case "删除不存在的帖子" "POST" "/post/delete" "{\"uid\": $VALID_UID,
\"token\": \"$VALID_TOKEN\", \"post_id\": 999999}" "1016"

    # 5.5 无权限操作（非作者删除帖子）
    # 先创建一个帖子
    log_test "创建测试帖子用于权限测试..."
    CREATE_RESPONSE=$(curl -s -X POST "$SERVER/post/create" \
        -H "Content-Type: application/json" \
        -d "{\"uid\": $VALID_UID, \"token\": \"$VALID_TOKEN\", \"forum_id\": 1, \"title\": \"
权限测试帖子\", \"content\": \"用于测试权限\"}")

    TEST_POST_ID=$(echo "$CREATE_RESPONSE" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    print(data.get('post_id', 0))
except:
    print(0)
")

    if [ "$TEST_POST_ID" -gt 0 ]; then
        log_test "创建了测试帖子 ID: $TEST_POST_ID"

        # 使用另一个"用户"尝试删除（模拟无权限）
        test_error_case "无权限删除帖子" "POST" "/post/delete" "{\"uid\": 999999, \"token\":
\"invalid_token\", \"post_id\": $TEST_POST_ID}" "1010"

        # 清理：删除测试帖子
        curl -s -X POST "$SERVER/post/delete" \

```

```

        -H "Content-Type: application/json" \
        -d "{\"uid\": $VALID_UID, \"token\": \"\$VALID_TOKEN\", \"post_id\":
$TEST_POST_ID}" > /dev/null
    fi

    # 5.6 置顶/加精权限测试（非贴吧 owner）
    # 先获取一个帖子 ID
    POSTS_RESPONSE=$(curl -s "$SERVER/posts?forum_id=1&page=1&limit=1")
    SAMPLE_POST_ID=$(echo "$POSTS_RESPONSE" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    posts = data.get('posts', [])
    if posts:
        print(posts[0].get('post_id', 0))
    else:
        print(0)
except:
    print(0)
")

    if [ "$SAMPLE_POST_ID" -gt 0 ]; then
        test_error_case "非贴吧 owner 置顶" "POST" "/set_top_post" "{\"uid\": $VALID_UID,
\"token\": \"\$VALID_TOKEN\", \"post_id\": $SAMPLE_POST_ID, \"is_top\": true}" "1017"
        test_error_case "非贴吧 owner 加精" "POST" "/set_essence_post" "{\"uid\": $VALID_UID,
\"token\": \"\$VALID_TOKEN\", \"post_id\": $SAMPLE_POST_ID, \"is_essence\": true}" "1017"
    fi
fi

echo ""
echo "6. 测试回复相关错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

if [ -n "$VALID_TOKEN" ]; then
    # 6.1 回复不存在的帖子
    test_error_case "回复不存在的帖子" "POST" "/reply" "{\"uid\": $VALID_UID, \"token\":
\"$VALID_TOKEN\", \"post_id\": 999999, \"content\": \"测试回复\", \"parent_reply_id\": 0}"
"1016"

    # 6.2 回复不存在的父回复
    if [ "$SAMPLE_POST_ID" -gt 0 ]; then
        test_error_case "回复不存在的父回复" "POST" "/reply" "{\"uid\": $VALID_UID,
\"token\": \"\$VALID_TOKEN\", \"post_id\": $SAMPLE_POST_ID, \"content\": \"测试回复\",
\"parent_reply_id\": 999}" "1015"
    fi
fi

```

```

# 6.3 删除不存在的回复
test_error_case "删除不存在的回复" "POST" "/reply/delete" '{"uid\: $VALID_UID,
\"token\: \"$VALID_TOKEN\", \"reply_id\: 150}' "1022"
fi

echo ""
echo "7. 测试搜索功能错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 7.1 空关键词搜索
test_error_case "空关键词搜索" "GET" "/search?keyword=&type=post&page=1&limit=10" ""
"1001" # 返回错误

# 7.2 无效的搜索类型
test_error_case "无效搜索类型" "GET" "/search?keyword=test&type=invalid&page=1&limit=10" ""
"1001" # 返回错误

echo ""
echo "8. 测试用户内容管理错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 8.1 无效 Token 查看用户内容
test_error_case "无效 Token 查看我的帖子" "GET"
"/my/posts?uid=1001&token=invalid_token&page=1&limit=10" "" "1010"

# 8.2 无权限查看他人内容（如果能测的话）
# 这个可能需要具体实现支持

echo ""
echo "9. 测试注册相关错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE

# 9.1 用户已存在 (1005)
# 需要知道一个已存在的用户名和邮箱
test_error_case "用户名已存在" "POST" "/user_register" '{"email": "secondtonone1@163.com",
"user": "tc", "passwd": "123456", "confirm": "123456", "icon": "", "varifypcode": "123456"}'
"1003"

test_error_case "邮箱已存在" "POST" "/user_register" '{"email": "18165031775@qq.com",
"user": "newusersss", "passwd": "123456", "confirm": "123456", "icon": "", "varifypcode":
"123456"}' "1003"

# 9.2 两次密码不一致 (1006)

```

```
test_error_case "密码不一致" "POST" "/user_register" '{"email": "testregister@example.com",
"user": "testregister", "passwd": "123456", "confirm": "654321", "icon": "", "varifycode":
"123456"}' "1006"
```

9.3 验证码错误 (1004)

```
test_error_case "验证码错误" "POST" "/user_register" '{"email": "testregister@example.com",
"user": "testregister", "passwd": "123456", "confirm": "123456", "icon": "", "varifycode":
"wrongcode"}' "1003"
```

9.4 验证码过期 (1003)

需要有一个过期的验证码，暂时跳过

```
echo ""
echo "10. 边界值和异常输入测试" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE
```

10.1 超长输入

```
LONG_STRING=$(printf '%*s' 10000 | tr ' ' 'x')
test_error_case "超长标题" "POST" "/post/create" '{"uid": $VALID_UID, "token":
"$VALID_TOKEN", "forum_id": 1, "title": "$LONG_STRING", "content": "测试内容"}'
"1015" # 可能成功或失败
```

10.2 SQL 注入尝试 (应该被参数化查询阻止)

```
test_error_case "SQL 注入标题" "POST" "/post/create" '{"uid": $VALID_UID, "token":
"$VALID_TOKEN", "forum_id": 1, "title": "test'); DROP TABLE forum; --",
"content": "测试内容"}' "0"
```

10.3 XSS 尝试

```
test_error_case "XSS 内容" "POST" "/post/create" '{"uid": $VALID_UID, "token":
"$VALID_TOKEN", "forum_id": 1, "title": "测试标题", "content":
"<script>alert('xss')</script>}" "0"
```

10.4 特殊字符

```
test_error_case "特殊字符" "POST" "/post/create" '{"uid": $VALID_UID, "token":
"$VALID_TOKEN", "forum_id": 1, "title": "测试!@#%^&*()标题", "content": "内
容!@#%^&*()}" "0"
```

```
echo ""
echo "11. 测试分页参数错误" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE
```

11.1 无效的分页参数

```
test_error_case "负数的页码" "GET" "/forums?page=-1&limit=10" "" "0"
test_error_case "零页码" "GET" "/forums?page=0&limit=10" "" "0"
test_error_case "负数的每页数量" "GET" "/forums?page=1&limit=-1" "" "0"
```

```
test_error_case "零每页数量" "GET" "/forums?page=1&limit=0" "" "0"
test_error_case "超大每页数量" "GET" "/forums?page=1&limit=10000" "" "0"
```

11.2 非数字参数

```
test_error_case "非数字页码" "GET" "/forums?page=abc&limit=10" "" "1001"
test_error_case "非数字每页数量" "GET" "/forums?page=1&limit=abc" "" "1001"
```

```
echo ""
echo "=== 测试总结 ===" | tee -a $LOG_FILE
echo "详细日志已保存到: $LOG_FILE" | tee -a $LOG_FILE
echo "===== " | tee -a $LOG_FILE
```

显示错误码统计

```
echo ""
echo "📊 测试的错误码分布:"
grep -o '"error":[0-9]*' "$LOG_FILE" | sed 's/"error"://' | sort | uniq -c | sort -rn | tee
-a $LOG_FILE
```

```
echo ""
echo "建议:"
echo "1. 检查未覆盖的错误码"
echo "2. 重点关注返回码不是期望值的测试用例"
echo "3. 查看日志文件获取详细响应信息"
```

得到的结果存于：error_test_20251203_001828.log

=== HTTP 接口错误码测试 ===

测试时间：Wed Dec 3 12:18:28 AM CST 2025

=====

1. 测试 JSON 解析错误 (1001)

=====

[00:18:28] 测试：无效 JSON 格式 (期望：1001)

[00:18:28] 响应：{

 "error" : 1001

}

[00:18:28] ☒ 通过：期望 1001，实际 1001

[00:18:28] 测试：缺少引号 (期望：1001)

[00:18:28] 响应：{

 "error" : 1001

}

[00:18:28] ☒ 通过：期望 1001，实际 1001

[00:18:28] 测试：多余逗号 (期望：1001)

[00:18:28] 响应：{

 "error" : 1001

}

[00:18:28] ☒ 通过：期望 1001，实际 1001

2. 测试用户登录相关错误

=====

[00:18:28] 测试：不存在的邮箱 (期望：1009)

[00:18:28] 响应：{

 "error" : 1009

}

[00:18:28] ☒ 通过：期望 1009，实际 1009

[00:18:28] 测试：错误密码 (期望：1009)

[00:18:29] 响应：{

 "error" : 1009

}

[00:18:29] ☒ 通过：期望 1009，实际 1009

[00:18:29] 测试：缺少 email (期望：1009)

[00:18:29] 响应：{

 "error" : 1009

}

[00:18:29] ☒ 通过：期望 1009，实际 1009

[00:18:29] 测试：缺少 password (期望：1009)

[00:18:29] 响应：{

 "error" : 1009

}

[00:18:29] ☒ 通过：期望 1009，实际 1009

[00:18:29] 测试: 空邮箱 (期望: 1009)

[00:18:29] 响应: {

"error" : 1009

}

[00:18:29] ☒ 通过: 期望 1009, 实际 1009

[00:18:29] 测试: 空密码 (期望: 1009)

[00:18:29] 响应: {

"error" : 1009

}

[00:18:29] ☒ 通过: 期望 1009, 实际 1009

3. 测试贴吧相关错误

=====

[00:18:29] 登录获取有效 token...

[00:18:29] 获取到有效 token: 417404f8-08a6-41ef-9...

[00:18:29] 测试: 不存在的贴吧 ID (期望: 1012)

[00:18:30] 响应: {

"error" : 1012

}

[00:18:30] ☒ 通过: 期望 1012, 实际 1012

[00:18:30] 测试: 非数字贴吧 ID (期望: 1001)

[00:18:30] 响应: {

"error" : 1001

}

[00:18:30] ☒ 通过: 期望 1001, 实际 1001

[00:18:30] 测试: 负数贴吧 ID (期望: 1012)

[00:18:30] 响应: {

"error" : 1012

}

[00:18:30] ☒ 通过: 期望 1012, 实际 1012

[00:18:30] 测试重复关注...

[00:18:30] 测试: 重复关注 (期望: 1013)

[00:18:30] 响应: {

"error" : 1013

}

[00:18:30] ☒ 通过: 期望 1013, 实际 1013

[00:18:30] 测试: 关注不存在的贴吧 (期望: 1012)

[00:18:30] 响应: {

"error" : 1012

}

[00:18:30] ☒ 通过: 期望 1012, 实际 1012

[00:18:30] 测试: 取消未关注的贴吧 (期望: 1014)

[00:18:30] 响应: {

"error" : 1014

}

[00:18:30] ☒ 通过: 期望 1014, 实际 1014

4. 测试 Token 相关错误 (1010)

=====

[00:18:30] 测试: 无效 Token (期望: 1010)

[00:18:31] 响应: {

 "error" : 1010

}

[00:18:31] ☒ 通过: 期望 1010, 实际 1010

[00:18:31] 测试: 空 Token (期望: 1001)

[00:18:31] 响应: {

 "error" : 1001

}

[00:18:31] ☒ 通过: 期望 1001, 实际 1001

[00:18:31] 测试: Token 与 UID 不匹配 (期望: 1010)

[00:18:31] 响应: {

 "error" : 1010

}

[00:18:31] ☒ 通过: 期望 1010, 实际 1010

5. 测试帖子相关错误

=====

[00:18:31] 测试: 不存在的帖子 (期望: 1016)

[00:18:31] 响应: {

 "error" : 1016

}

[00:18:31] ☒ 通过: 期望 1016, 实际 1016

[00:18:31] 测试: 在不存在贴吧发帖 (期望: 1012)

[00:18:31] 响应: {

 "error" : 1012

}

[00:18:31] ☒ 通过: 期望 1012, 实际 1012

[00:18:31] 测试: 更新不存在的帖子 (期望: 1016)

[00:18:31] 响应: {

 "error" : 1016

}

[00:18:31] ☒ 通过: 期望 1016, 实际 1016

[00:18:31] 测试: 删除不存在的帖子 (期望: 1016)

[00:18:32] 响应: {

 "error" : 1016

}

[00:18:32] ☒ 通过: 期望 1016, 实际 1016

[00:18:32] 创建测试帖子用于权限测试...

[00:18:32] 创建了测试帖子 ID: 31

[00:18:32] 测试: 无权限删除帖子 (期望: 1010)

[00:18:32] 响应: {

 "error" : 1010


```
}
[00:18:32] ☒ 通过: 期望 1010, 实际 1010
[00:18:32] 测试: 非贴吧 owner 置顶 (期望: 1017)
[00:18:32] 响应: {
  "error" : 1017
}
[00:18:32] ☒ 通过: 期望 1017, 实际 1017
[00:18:32] 测试: 非贴吧 owner 加精 (期望: 1017)
[00:18:32] 响应: {
  "error" : 1017
}
[00:18:32] ☒ 通过: 期望 1017, 实际 1017
```

6. 测试回复相关错误

```
=====
[00:18:32] 测试: 回复不存在的帖子 (期望: 1016)
[00:18:32] 响应: {
  "error" : 1016
}
[00:18:32] ☒ 通过: 期望 1016, 实际 1016
[00:18:32] 测试: 回复不存在的父回复 (期望: 1015)
[00:18:33] 响应: {
  "error" : 1015
}
[00:18:33] ☒ 通过: 期望 1015, 实际 1015
[00:18:33] 测试: 删除不存在的回复 (期望: 1022)
[00:18:33] 响应: {
  "error" : 1022
}
[00:18:33] ☒ 通过: 期望 1022, 实际 1022
```

7. 测试搜索功能错误

```
=====
[00:18:33] 测试: 空关键词搜索 (期望: 1001)
[00:18:33] 响应: {
  "error" : 1001
}
[00:18:33] ☒ 通过: 期望 1001, 实际 1001
[00:18:33] 测试: 无效搜索类型 (期望: 1001)
[00:18:33] 响应: {
  "error" : 1001
}
[00:18:33] ☒ 通过: 期望 1001, 实际 1001
```

8. 测试用户内容管理错误

```
=====
[00:18:33] 测试: 无效 Token 查看我的帖子 (期望: 1010)
[00:18:33] 响应: {
```

```
    "error" : 1010
}
[00:18:33] ☒ 通过: 期望 1010, 实际 1010
```

9. 测试注册相关错误

```
=====
[00:18:33] 测试: 用户名已存在 (期望: 1003)
[00:18:33] 响应: {
    "error" : 1003
}
[00:18:33] ☒ 通过: 期望 1003, 实际 1003
[00:18:33] 测试: 邮箱已存在 (期望: 1003)
[00:18:33] 响应: {
    "error" : 1003
}
[00:18:33] ☒ 通过: 期望 1003, 实际 1003
[00:18:33] 测试: 密码不一致 (期望: 1006)
[00:18:33] 响应: {
    "error" : 1006
}
[00:18:33] ☒ 通过: 期望 1006, 实际 1006
[00:18:33] 测试: 验证码错误 (期望: 1003)
[00:18:33] 响应: {
    "error" : 1003
}
[00:18:33] ☒ 通过: 期望 1003, 实际 1003
```

10. 边界值和异常输入测试

```
=====
[00:18:33] 测试: 超长标题 (期望: 1015)
[00:18:34] 响应: {
    "error" : 1015
}
[00:18:34] ☒ 通过: 期望 1015, 实际 1015
[00:18:34] 测试: SQL 注入标题 (期望: 0)
[00:18:34] 响应: {
    "error" : 0,
    "post_id" : 32
}
[00:18:34] ☒ 通过: 期望 0, 实际 0
[00:18:34] 测试: XSS 内容 (期望: 0)
[00:18:34] 响应: {
    "error" : 0,
    "post_id" : 33
}
[00:18:34] ☒ 通过: 期望 0, 实际 0
[00:18:34] 测试: 特殊字符 (期望: 0)
```

[00:18:34] 响应: {

 "error" : 0,
 "post_id" : 34

}

[00:18:34] ☒ 通过: 期望 0, 实际 0

11. 测试分页参数错误

=====

[00:18:34] 测试: 负数的页码 (期望: 0)

[00:18:34] 响应: {

 "error" : 0,

 "forums" :

[

{

 "avatar" : "/res/forum_tech.jpg",
 "description" : "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
 "follower_count" : 2,
 "forum_id" : 1,
 "name" : "\u79d1\u6280\u8ba8\u8bba\u5427",
 "post_count" : 25

},

{

 "avatar" : "/res/forum_photo.jpg",
 "description" : "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
 "follower_count" : 2,
 "forum_id" : 2,
 "name" : "\u6444\u5f71\u4ea4\u6d41\u5427",
 "post_count" : 2

},

{

 "avatar" : "/res/forum_game.jpg",
 "description" : "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
 "follower_count" : 1,
 "forum_id" : 3,
 "name" : "\u6e38\u620f\u7ec4\u961f\u5427",
 "post_count" : 1

}

]

}

[00:18:34] ☒ 通过: 期望 0, 实际 0

[00:18:34] 测试: 零页码 (期望: 0)

[00:18:34] 响应: {

 "error" : 0,

 "forums" :

[

{

```

    "avatar" : "/res/forum_tech.jpg",
    "description" : "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
    "follower_count" : 2,
    "forum_id" : 1,
    "name" : "\u79d1\u6280\u8ba8\u8bba\u5427",
    "post_count" : 25
  },
  {
    "avatar" : "/res/forum_photo.jpg",
    "description" : "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
    "follower_count" : 2,
    "forum_id" : 2,
    "name" : "\u6444\u5f71\u4ea4\u6d41\u5427",
    "post_count" : 2
  },
  {
    "avatar" : "/res/forum_game.jpg",
    "description" : "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
    "follower_count" : 1,
    "forum_id" : 3,
    "name" : "\u6e38\u620f\u7ec4\u961f\u5427",
    "post_count" : 1
  }
]
}

```

[00:18:34] ☒ 通过: 期望 0, 实际 0

[00:18:34] 测试: 负数的每页数量 (期望: 0)

[00:18:34] 响应: {

```

  "error" : 0,
  "forums" :
  [
    {
      "avatar" : "/res/forum_tech.jpg",
      "description" : "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
      "follower_count" : 2,
      "forum_id" : 1,
      "name" : "\u79d1\u6280\u8ba8\u8bba\u5427",
      "post_count" : 25
    },
    {
      "avatar" : "/res/forum_photo.jpg",
      "description" : "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
      "follower_count" : 2,
      "forum_id" : 2,
      "name" : "\u6444\u5f71\u4ea4\u6d41\u5427",

```

```

        "post_count" : 2
    },
    {
        "avatar" : "/res/forum_game.jpg",
        "description" : "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
        "follower_count" : 1,
        "forum_id" : 3,
        "name" : "\u6e38\u620f\u7ec4\u961f\u5427",
        "post_count" : 1
    }
]
}

[00:18:34] ☒ 通过: 期望 0, 实际 0
[00:18:34] 测试: 零每页数量 (期望: 0)
[00:18:34] 响应: {
    "error" : 0,
    "forums" :
    [
        {
            "avatar" : "/res/forum_tech.jpg",
            "description" : "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
            "follower_count" : 2,
            "forum_id" : 1,
            "name" : "\u79d1\u6280\u8ba8\u8bba\u5427",
            "post_count" : 25
        },
        {
            "avatar" : "/res/forum_photo.jpg",
            "description" : "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
            "follower_count" : 2,
            "forum_id" : 2,
            "name" : "\u6444\u5f71\u4ea4\u6d41\u5427",
            "post_count" : 2
        },
        {
            "avatar" : "/res/forum_game.jpg",
            "description" : "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
            "follower_count" : 1,
            "forum_id" : 3,
            "name" : "\u6e38\u620f\u7ec4\u961f\u5427",
            "post_count" : 1
        }
    ]
}

```

[00:18:34] ☒ 通过: 期望 0, 实际 0

[00:18:35] 测试: 超大每页数量 (期望: 0)

[00:18:35] 响应: {

```
"error" : 0,
"forums" :
[
  {
    "avatar" : "/res/forum_tech.jpg",
    "description" : "\u804a\u786c\u4ef6/AI/\u540e\u7aef",
    "follower_count" : 2,
    "forum_id" : 1,
    "name" : "\u79d1\u6280\u8ba8\u8bba\u5427",
    "post_count" : 25
  },
  {
    "avatar" : "/res/forum_photo.jpg",
    "description" : "\u5668\u6750\u3001\u540e\u671f\u4e0e\u65c5\u62cd\u5206\u4eab",
    "follower_count" : 2,
    "forum_id" : 2,
    "name" : "\u6444\u5f71\u4ea4\u6d41\u5427",
    "post_count" : 2
  },
  {
    "avatar" : "/res/forum_game.jpg",
    "description" : "\u5f00\u9ed1\u3001\u653b\u7565\u4e0e\u8054\u673a\u7ea6\u6218",
    "follower_count" : 1,
    "forum_id" : 3,
    "name" : "\u6e38\u620f\u7ec4\u961f\u5427",
    "post_count" : 1
  }
]
```

[00:18:35] ☒ 通过: 期望 0, 实际 0

[00:18:35] 测试: 非数字页码 (期望: 1001)

[00:18:35] 响应: {

```
"error" : 1001
```

}

[00:18:35] ☒ 通过: 期望 1001, 实际 1001

[00:18:35] 测试: 非数字每页数量 (期望: 1001)

[00:18:35] 响应: {

```
"error" : 1001
```

}

[00:18:35] ☒ 通过: 期望 1001, 实际 1001

=== 测试总结 ===

详细日志已保存到: [error_test_20251203_001828.log](#)

=====